

Unconfoundedness II: Propensity Scores and Regression

BUSS975: Causal Inference in Financial Research

Professor Ji-Woong Chung

Korea University

Outline

1. Propensity scores
2. Regression
3. The NSW demonstration

1. Propensity Scores

Where Lecture 3a left us

- Unconfoundedness says: **compare treated and controls with the same X .**
- **Subclassification** and **matching** suffer from the curse of dimensionality.
- Must we match on all of X ? The assumption is stated conditional on X , so it seems so.
- **Rosenbaum and Rubin (1983): no.** One scalar computed from X does the whole job.

Defining the propensity score

- **Propensity score:** $e(X) = \Pr(D = 1 \mid X)$
 - ▶ “the conditional probability of assignment to a particular treatment given a vector of observed covariates” (Rosenbaum and Rubin 1983)
- Three things it is:
 - ▶ A **probability**, so it lives in $[0, 1]$.
 - ▶ A **conditional** probability — read it as: *of the units that look like this on X , what fraction are treated?*
 - ▶ A description of the **assignment mechanism**, not of the outcome.
- Two results follow: the **balancing property** (conditional on $e(X)$, treated and control look alike on X) and the **propensity score theorem** (conditioning on $e(X)$ does what conditioning on all of X does).

The balancing property

Claim. $D \perp\!\!\!\perp X \mid e(X)$

D is binary, so its distribution is summarised by its mean; the claim is equivalent to $E[D \mid X, e(X)] = E[D \mid e(X)]$. We show **both sides equal** $e(X)$, and hence each other.

Left side. $e(X)$ is a function of X , so conditioning on it adds nothing to X :

$$E[D \mid X, e(X)] = E[D \mid X] = e(X) \quad (\text{definition of } e(X)).$$

Right side. Apply the **law of iterated expectations**, $E[A \mid C] = E_B[E[A \mid B, C] \mid C]$, with $A = D$, $B = X$, $C = e(X)$:

$$E[D \mid e(X)] = E_X \left[\underbrace{E[D \mid X, e(X)]}_{= e(X) \text{ by the left side}} \mid e(X) \right] = E_X[e(X) \mid e(X)] = e(X). \blacksquare$$

- The mean version, $E[X \mid D=1, e(X)] = E[X \mid D=0, e(X)]$, is what a balance table tests.

The propensity score theorem

Claim. If $(Y^1, Y^0) \perp\!\!\!\perp D \mid X$, then $(Y^1, Y^0) \perp\!\!\!\perp D \mid e(X)$

$$\text{In mean form, } \underbrace{E[D \mid Y^1, Y^0, X] = E[D \mid X]}_{\text{assumed}} \implies \underbrace{E[D \mid Y^1, Y^0, e(X)] = E[D \mid e(X)]}_{\text{to show}}.$$

The right-hand side is $e(X)$ by the previous slide, so only the left-hand side is left to prove:

$$\begin{aligned} E[D \mid Y^1, Y^0, e(X)] &= E_X \left[E[D \mid Y^1, Y^0, X, e(X)] \mid Y^1, Y^0, e(X) \right] && \text{law of iterated exp.} \\ &= E_X \left[E[D \mid Y^1, Y^0, X] \mid Y^1, Y^0, e(X) \right] && e(X) \text{ is a function of } X \\ &= E_X \left[\underbrace{E[D \mid X]}_{= e(X)} \mid Y^1, Y^0, e(X) \right] = e(X). \blacksquare && \text{unconfoundedness} \end{aligned}$$

- The *ATE* is identified by comparing treated and control units at the same **score** rather than the same X : $ATE = E \left[E[Y \mid D=1, e(X)] - E[Y \mid D=0, e(X)] \right]$

Estimating the score

- You know the **true** score only if you ran the experiment. Otherwise estimate it: $\Pr(D_i=1 \mid X_i) = F(\beta X_i)$, by **logit or probit** — not OLS, which extrapolates linearly and will hand you “probabilities” outside $[0, 1]$.
- Judge the specification by **balance achieved**, not by fit. A score with a high pseudo- R^2 but poor balance is a bad score; the reverse is fine.
- Fit, check balance, add interactions and polynomials where balance fails, refit. A legitimate specification search: it never looks at Y , so the stage is **blind to the outcome**. Say so in the paper.
- **The score creates no support that X lacks.** If a cell is all-treated or all-control, collapsing X to a scalar hides that; it does not fix it.

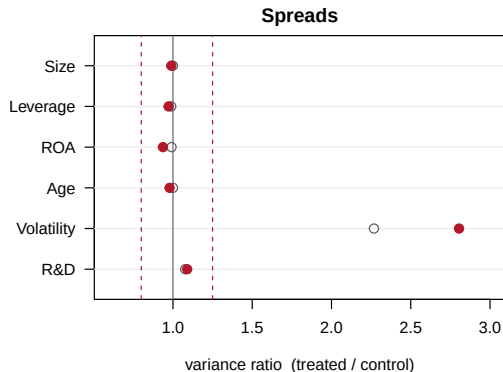
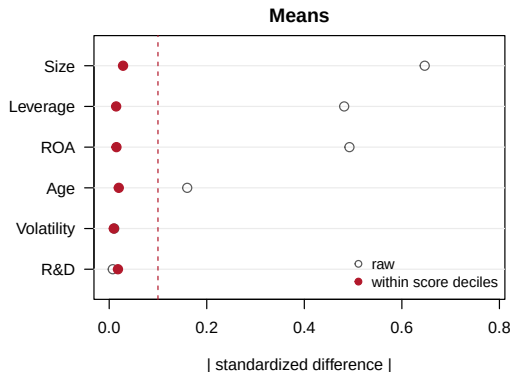
Checking balance: not a t -test

The balancing property is checkable. Report it covariate by covariate, before and after.

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{(s_1^2 + s_0^2)/2}}, \quad \text{VR} = \frac{s_1^2}{s_0^2}$$

- **Standardized mean difference.** $|\text{SMD}| < 0.1$ is good.
 - ▶ Keep the denominator **fixed at its raw-sample value** when recomputing after adjustment, or the yardstick moves with the thing you are measuring.
- **Variance ratio.** Means can match while spreads do not. Rubin(2001): stay inside $[0.5, 2]$, ideally $[0.8, 1.25]$.
- **Do not use t -tests.** Balance is a property of *this sample*, not a population hypothesis — and p confounds imbalance with sample size (Imai, King and Stuart 2008).

A balance plot: means are not enough



Selection here is linear in Size, Leverage, ROA, Age — and **quadratic in Volatility**. The score was fitted with linear terms only.

Reading the balance plot

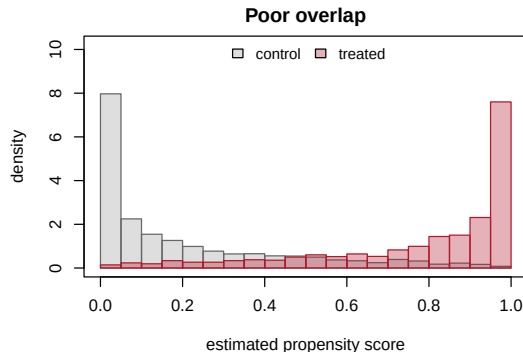
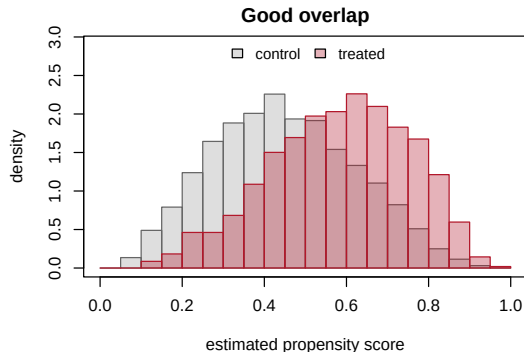
- **Left panel, the success.** Four covariates start badly imbalanced — Size at 0.65, Leverage and ROA near 0.49. Compared *within deciles of the score* all fall under 0.03. This is the plot referees want, and where most papers stop.
- **Left panel, the trap.** Volatility looks *perfect* from the start: $SMD = -0.010$.
- **Right panel.** Its variance ratio is **2.27** — treated firms have more than twice the spread. Extreme volatility drives selection in *both* tails.
- **Conditioning on the score makes it worse**, $2.27 \rightarrow 2.80$, because the fitted score never contained vol^2 . Add that one term and the ratio falls to **1.07**.
- Judge the score by **balance achieved, not by fit** — and balance means the whole distribution. Check squares and interactions, not just levels.

Three things to do with the score

Once every unit has a $\hat{e}(X_i)$, three standard uses:

1. **Diagnose.** Plot $\hat{e}$ separately for treated and control units. This is the common-support check, and it costs nothing.
 2. **Weight.** Reweight the sample to build a pseudo-experiment — inverse probability weighting.
 3. **Match.** Pair treated to control units with similar scores, then difference — propensity score matching.
- Busso, DiNardo and McCrary (2014): reweighting is competitive with the best matching estimators when overlap is good, whereas matching (e.g. nearest neighbour with bias correction) can be more effective when overlap is poor.

Visualizing overlap



Right panel: a treated mass at $\hat{e} \approx 1$ with no controls, and a control mass at $\hat{e} \approx 0$ with no treated. Those units have **no counterfactual**. Any estimate covering them is extrapolation, not comparison.

This plot belongs in every such paper

- Common support is verifiable, unlike unconfoundedness.
- What to do when it looks like the right panel:
 - ▶ **Trim** extreme scores, or impose a caliper: drop units with $\hat{e} \notin [0.1, 0.9]$. A rule of thumb, but a defensible one (Crump, Hotz, Imbens and Mitnik, 2009).
 - ▶ **Report the change in estimand.** After trimming you describe a subpopulation. Say how many observations went and how the treated sample shifted.
 - ▶ Consider retreating to the *ATT*, which only needs overlap in one direction.
- What *not* to do: report the estimate and stay silent. Regression will extrapolate into a region with no data and give a standard error reflecting none of it.

Where the weighting idea comes from

- The idea predates causal inference. **Horvitz and Thompson (1952)**, from survey sampling.
- Among firms with $\hat{e} = 0.1$, only one in ten was treated. The treated arm under-represents this type tenfold, so each such treated firm stands in for ten: weight it by $1/\hat{e} = 10$. A treated firm with $\hat{e} = 0.9$ is typical of its type: weight $1/0.9 \approx 1.1$.
- Do this in **both arms** — treated by $1/\hat{e}$, control by $1/(1 - \hat{e})$ — and you build a **pseudo-population** in which treatment no longer depends on X . Selection is undone by *recounting*, not by discarding or stratifying.
- The weight depends on **which arm** a unit is in and on **which estimand** you want. And the danger is already visible: a score near 0 or 1 produces an enormous weight.

Inverse probability weighting

Create a balanced pseudo-population by inverse probability weighting. Write $\hat{e}_i$ for $\hat{e}(X_i)$ and $\pi = \Pr(D=1)$.

$$ATE = E[Y^1 - Y^0] = E\left[\frac{D Y}{e(X)}\right] - E\left[\frac{(1-D) Y}{1-e(X)}\right]$$

$$ATT = E[Y^1 - Y^0 | D=1] = \frac{1}{\pi} \left(E[DY] - E\left[\frac{e(X)(1-D)Y}{1-e(X)}\right] \right)$$

$$\widehat{ATE}_{IPW} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i}{\hat{e}_i} - \frac{1-D_i}{1-\hat{e}_i} \right] Y_i$$

$$\widehat{ATT}_{IPW} = \frac{1}{n_T} \sum_{i=1}^n \left[D_i - (1-D_i) \frac{\hat{e}_i}{1-\hat{e}_i} \right] Y_i$$

- Both are **Horvitz–Thompson (1952)** forms.

Why *that* weight for the ATE?

We prove $E\left[\frac{D \cdot Y}{e(X)}\right] = E[Y^1]$. The $E[Y^0]$ half is symmetric.

$$E\left[\frac{D \cdot Y}{e(X)}\right] = E\left[\frac{D \cdot Y^1}{e(X)}\right]$$

switching equation (consistency): $D \cdot Y = D \cdot Y^1$

$$= E_X\left[E\left(\frac{D \cdot Y^1}{e(X)} \mid X\right)\right]$$

law of total expectation

$$= E_X\left[\frac{1}{e(X)} \cdot E(D \cdot Y^1 \mid X)\right]$$

$e(X)$ is fixed given X

$$= E_X\left[\frac{1}{e(X)} \cdot E(D \mid X) \cdot E(Y^1 \mid X)\right]$$

unconfoundedness (exchangeability): $D \perp\!\!\!\perp Y^1 \mid X$

$$= E_X\left[\frac{1}{e(X)} \cdot e(X) \cdot E(Y^1 \mid X)\right]$$

by definition, $E(D \mid X) = e(X)$

$$= E_X[E(Y^1 \mid X)] = E[Y^1]$$

cancel; law of total expectation

Why *that* weight for the ATT?

$$E[Y^1 \mid D=1] = E[Y \mid D=1] = \frac{E[DY]}{E[D]}.$$

$$E[Y^0 \mid D=1] = E_X \left[E(Y^0 \mid X, D=1) \mid D=1 \right]$$

law of total expectation, over $f(x \mid D=1)$

$$= E_X \left[E(Y^0 \mid X, D=0) \mid D=1 \right]$$

unconfoundedness — swap the arm

$$= E_X \left[E(Y^0 \mid X, D=0) \cdot \frac{e(X)}{\pi} \right]$$

Bayes: $f(x \mid D=1) = e(x)f(x)/\pi$

$$= E_X \left[\frac{E[(1-D)Y \mid X]}{1 - e(X)} \cdot \frac{e(X)}{\pi} \right]$$

$$E[(1-D)Y \mid X] = (1 - e(X))E[Y^0 \mid X, D=0]$$

$$= E \left[\frac{(1-D)Y e(X)}{(1 - e(X)) \pi} \right]$$

law of total expectation

Line 3 converts an average over the *treated* covariate distribution into one over the *whole* population.

Ten banks, split by size

Lecture 1's capital injections, now with a covariate: bank **size**.

Bank	Size	D	Y^0	Y^1	τ	observed Y
1	Small	1	10	12	2	12
2	Small	0	8	10	2	8
3	Small	0	9	11	2	9
4	Small	0	11	13	2	11
5	Small	0	12	14	2	12
6	Large	1	18	24	6	24
7	Large	1	19	25	6	25
8	Large	1	21	27	6	27
9	Large	1	22	28	6	28
10	Large	0	20	26	6	20

- Small banks are **rarely** rescued, 1 of 5; large banks **usually** are, 4 of 5. So $e(\text{Small}) = 0.2$ and $e(\text{Large}) = 0.8$.
- Small banks gain 2, large banks gain 6. Truth: $ATE = 4$, $ATT = 5.2$.
- Only the **bold** column is observed. $SDO = 11.2$.

The ATE rebuilds the population

Cell	N	e	injected	$\times 1/e$	controls	$\times 1/(1 - e)$
Small	5	0.2	1	$\times 5 = \mathbf{5}$	4	$\times 1.25 = \mathbf{5}$
Large	5	0.8	4	$\times 1.25 = \mathbf{5}$	1	$\times 5 = \mathbf{5}$
Total	10		5	10	5	10

- Bank 1 is the **only** small bank rescued, so it must stand in for all five. Its weight is $1/0.2 = 5$.
- Every weighted entry comes back to 5: each arm is blown up to the **full population of ten**. That is the *pseudo-population*, and in it $D \perp\!\!\!\perp X$ by construction.
- Before weighting, small banks are 20% of the injected sample and 80% of controls. After weighting, 50% of both — the imbalance causing the bias is gone.

The *ATT* reweights controls only

The *ATT* wants the **treated** distribution. The four large and one small rescued banks are already a sample from it, so they keep weight 1. Only the controls move.

Cell	controls	$\times e/(1 - e)$	=	injected
Small	4	0.25	1	1
Large	1	4.00	4	4

- Multiplying the controls by $e/(1 - e)$ makes their count match the injected count **exactly** — last column.
- Bank 10 is the lone large control and must cover **four** rescued banks, so it carries weight 4.
- If e were 1 — all five large banks rescued, there is no large control to reweight and create counterfactuals: **this is where the *ATT* fails**, and why it needs only $e(X) < 1$ while the *ATE* needs $0 < e(X) < 1$.

Does it work?

$$\widehat{E[Y^1]} = \frac{1}{10} \left[\frac{12}{0.2} + \frac{24+25+27+28}{0.8} \right] = \frac{60+130}{10} = 19$$

$$\widehat{E[Y^0]} = \frac{1}{10} \left[\frac{8+9+11+12}{0.8} + \frac{20}{0.2} \right] = \frac{50+100}{10} = 15$$

$$\widehat{ATE} = \widehat{E[Y^1]} - \widehat{E[Y^0]} = 19 - 15 = 4$$

$$\widehat{ATT} = \underbrace{\frac{12+24+25+27+28}{5}}_{\text{injected mean} = 23.2} - \underbrace{\frac{(8+9+11+12)(0.25)+20(4)}{4(0.25)+1(4)}}_{\text{controls} \times \text{odds} = 18.0} = 5.2$$

- Truth: $ATE = 4$, $ATT = 5.2$. **Both exactly recovered.** The naive SDO is $23.2 - 12 = \mathbf{11.2}$, nearly three times the ATE .
- $ATT > ATE$ because large banks gain more, and large banks are where the injections went. **Selection on gains — the two estimands correctly disagree.**

Two ways to divide

Both estimators use the same weights $1/\hat{e}_i$. They differ **only in the denominator**.

$$\underbrace{\widehat{E}[Y^1]_{\text{HT}} = \frac{1}{N} \sum_i \frac{D_i Y_i}{\hat{e}_i}}_{\text{divide by } N}$$

$$\underbrace{\widehat{E}[Y^1]_{\text{Hájek}} = \frac{1}{\sum_i D_i / \hat{e}_i} \sum_i \frac{D_i Y_i}{\hat{e}_i}}_{\text{divide by the weights you got}}$$

- In the **population** they agree: $E[D/e(X)] = 1$, so the weights sum to N .
- In a **sample** they do not. $\sum_i D_i / \hat{e}_i$ is only an *estimate* of N , and it misses.
- Hájek's weights sum to one by construction, so it is always a genuine **weighted average of outcomes you actually observed**. HT carries no such guarantee.
- The default in Stata's `teffects ipw` and R's `WeightIt`.

Example 1

Four rescued banks. **Every one has lending growth** $Y = 10$, so the answer must be 10 for $\widehat{E[Y^1]}$. Here $N = 8$ (including control with $D = 0$).

Bank	$\hat{e}$	weight $1/\hat{e}$	Y
A	0.50	2	10
B	0.50	2	10
C	0.50	2	10
D	0.25	4	10
sum		10	

$$\text{HT} = \frac{20 + 20 + 20 + 40}{8} = \frac{100}{8} = \mathbf{12.5}, \quad \text{Hájek} = \frac{100}{10} = \mathbf{10}.$$

- The weights sum to 10, not 8. HT divides by 8 anyway, returning $10 \times \frac{10}{8}$.

Example 2

Bank	$\hat{e}$	weight	share of weight	Y
A	0.50	2	3.6%	10
B	0.50	2	3.6%	10
C	0.50	2	3.6%	10
D	0.02	50	89.3%	24
sum		56	100%	

$$\text{HT} = \frac{1260}{8} = \mathbf{157.5}, \quad \text{Hájek} = \frac{1260}{56} = \mathbf{22.5}.$$

- D carries **89%** of the weight.
- **Normalization is necessary, not sufficient.** Check overlap separately: trim, or retreat to the *ATT* when the trouble is at $\hat{e} \approx 0$.

Propensity score matching

- Match each treated unit to controls with a **similar score**, then difference.
 - ▶ **Caliper** matching: refuse any match farther than some tolerance (say 0.001 on the score), *discarding* the treated unit if none qualifies.
 - ▶ **Radius** matching: average all controls inside the tolerance.
- Two problems with both:
 - ▶ The tolerance is **chosen by the researcher**
 - ▶ Dropping treated units **changes the estimand**: the *ATT* summarised the *original* treated sample.
- King and Nielsen (2019): two units can share a score, $e(X_i) = e(X_j)$, with opposite covariate profiles $X_i \neq X_j$. Matching on the scalar score can *increase* local imbalance relative to multivariate distance matching.

What the score does and does not buy

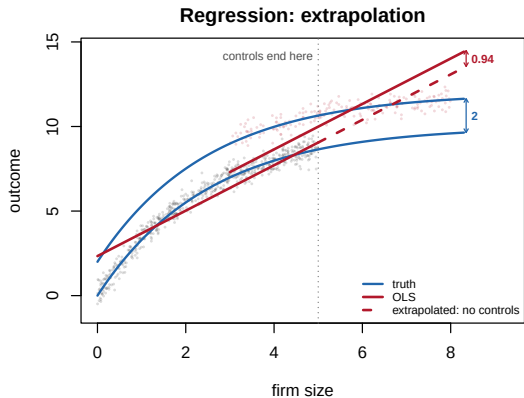
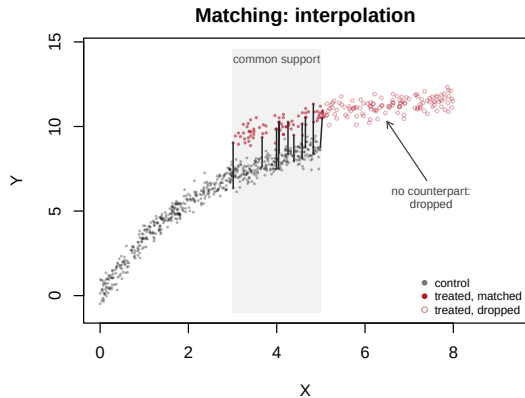
- It buys **dimension reduction**, not identification. If X omits a confounder, $e(X)$ omits it too.
- Regression hides missing common support behind linear extrapolation; plotting $\hat{e}(X)$ exposes it.
- A high pseudo- R^2 means covariates nearly predict treatment ($e(X) \rightarrow 0$ or 1): positivity breaks and IPW variance explodes. A low one means treatment is nearly independent of X — close to an RCT on observables.
- IPW shifts the modelling assumption from the outcome equation $E[Y \mid D, X]$ to the assignment equation $P(D=1 \mid X)$.

2. Regression

Crossing to the other side of the map

- Everything so far has been an **interpolation** method: subclassification, matching, the propensity score. Common support is the binding constraint.
- Regression sits on the other side: it needs no common support, so **the curse of dimensionality does not arise**.
- It buys that with a **functional form**: OLS imputes the counterfactual by **extrapolating** into regions with no comparable units.
- The trade: matching needs overlap; regression needs a correctly specified outcome model.
- Both still need unconfoundedness. Nothing in this section relaxes it.

Matching vs. regression



What exactly is that coefficient?

- In practice you will estimate the effect by running

$$Y_i = \alpha + \delta D_i + \gamma X_i + \varepsilon_i.$$

- What *is* $\hat{\delta}$? It is generally **neither the ATE nor the ATT**.
- Two questions come first:
 1. **Which variation in D** does OLS use, once X is in the regression?
 2. **How does it weight** the units that supply that variation?
- **Frisch–Waugh–Lovell** answers the first: OLS uses only the part of D that X cannot predict. The weighting result then follows from it.

Coefficients as scaled covariances

- With one regressor,

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad \hat{\beta}_1 = \frac{\widehat{\text{Cov}}(Y_i, X_i)}{\widehat{\text{Var}}(X_i)}.$$

- The coefficient is the slope of the best-fit line: a **covariance scaled by a variance**.
- Now add covariates:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_K X_{Ki} + \varepsilon_i.$$

- $\hat{\beta}_1$ is no longer $\widehat{\text{Cov}}(Y, X_1)/\widehat{\text{Var}}(X_1)$. **Frisch and Waugh (1933); Lovell (1963)**: it is still a scaled covariance — of *residualized* variables.

Frisch–Waugh–Lovell in three steps

Step 1. Auxiliary regression:

$$X_{1i} = \gamma_0 + \gamma_2 X_{2i} + \cdots + \gamma_K X_{Ki} + f_i.$$

$\hat{X}_{1i}$ is the part of X_{1i} the other covariates explain.

Step 2. Residualization:

$$\tilde{X}_{1i} = X_{1i} - \hat{X}_{1i}.$$

The variation in X_{1i} that is *uncorrelated* with everything else — **partialling out**.

Step 3. Regress the outcome on the residual.

$$Y_i = \alpha + \beta_1 \tilde{X}_{1i} + \varepsilon_i, \quad \hat{\beta}_1 = \frac{\widehat{\text{Cov}}(Y_i, \tilde{X}_{1i})}{\widehat{\text{Var}}(\tilde{X}_{1i})}.$$

- $\hat{\beta}_1$ is **identical** to the coefficient on X_{1i} in the full regression.

FWL is an identity, not an approximation

Simulation: 4,000 workers, earnings on a treatment indicator d , age, and high-school GPA, with a true effect of \$1,500.

	(1) Multivariate OLS	(2) FWL
d	1503.201 (8.249)	1503.201 (8.287)
age	1.173 (2.023)	—
gpa	9.081 (4.615)	—
N	4000	4000

- The estimate sits \$3.20 above the true \$1,500 — 0.39 standard errors, i.e. sampling noise.
- The standard errors do *not* agree, and that is not sampling noise.

Residualize both sides

- Another version residualizes the outcome too:

$$\begin{aligned}\tilde{Y}_i &= Y_i - \hat{Y}_i \quad \text{from} \quad Y_i = \lambda_0 + \lambda_2 X_{2i} + \cdots + \lambda_K X_{Ki} + v_i, \\ \text{then} \quad \tilde{Y}_i &= \beta_1 \tilde{X}_{1i} + \varepsilon_i.\end{aligned}$$

- Both versions return **exactly the same** $\hat{\beta}_1$.
- The double-residual version buys one more thing: **its residuals are those of the full regression.**

Three regressions, one coefficient

	(1) full OLS	(2) single, $Y \sim \tilde{D}$	(3) double, $\tilde{Y} \sim \tilde{D}$
<i>original</i> : the controls explain 0.1% of earnings			
d	1503.201 (8.249)	1503.201 (8.287)	1503.201 (8.247)
<i>expanded</i> : the controls explain 93%			
d	1503.201 (8.249)	1503.201 (91.493)	1503.201 (8.247)

- (1) and (3) differ only in degrees of freedom ($n - k - 1$ vs. $n - 2$); the residuals are identical.
- (2) residualizes only D . The part of Y that age and GPA explain goes to the error term. Harmless at $R^2 = 0.1\%$; fatal at 93%.

Why FWL is worth knowing

- **It tells you what variation identifies your coefficient.** $\hat{\beta}_1$ uses only the part of X_1 that the controls cannot explain.
- **It makes the estimate visualizable.** Residualize *both* sides and scatter $\tilde{Y}$ against $\tilde{X}_1$: a multivariate coefficient becomes a picture whose slope is $\hat{\beta}_1$.
- **It is the engine behind much else.** Fixed effects, double machine learning and the weighting results below are all FWL in disguise.
- Angrist and Pischke (2009) call this **regression anatomy**, and it is the single most useful algebraic fact about OLS.

Exogeneity, and the canonical OLS model

- The **canonical OLS model**:

$$Y_i = \alpha + \delta D_i + \gamma X_i + \varepsilon_i.$$

- $E[\varepsilon_i \mid D_i, X_i] = 0$ means “ D is exogenous.” It bundles three assumptions:
 - ▶ **Unconfoundedness**: $E[Y^0 \mid D, X] = E[Y^0 \mid X]$.
 - ▶ **Linearity**: $E[Y^0 \mid X] = \alpha + \gamma X$.
 - ▶ **Constant effect**: $\tau_i = \delta$ for every i .
- We take X to be pre-treatment throughout. If D moves X , X is an outcome, not a control (Lecture 2).

When the three assumptions fail

Each failure pushes something into ε_i that is correlated with D_i .

- **Linearity fails:** $E[Y^0 | X] = g(X) \neq \alpha + \gamma X$. The unmodeled part $g(X) - \alpha - \gamma X$ enters ε_i and is correlated with D through X .
 - ▶ *Fix:* flexible controls — polynomials, splines, category dummies
- **Constant effect fails:** $\tau_i \neq \delta$. The term $D_i(\tau_i - \bar{\tau})$ enters ε_i . $\hat{\delta}_{OLS}$ becomes a variance-weighted average of τ_i , overweighting cells with high $\text{Var}(D | X)$, and can differ sharply from the ATE.
 - ▶ *Fix:* interact D_i with covariates
- **Unconfoundedness fails:** $E[Y^0 | D, X] \neq E[Y^0 | X]$, i.e. selection on unobservables: $E[Y^0 | D=1, X] \neq E[Y^0 | D=0, X]$.
 - ▶ *Fix:* change the design!

OLS as a variance-weighted average

Claim With X saturated and unconfoundedness, $\hat{\delta}_{OLS} \rightarrow E[\sigma_D^2(X) \tau(X)] / E[\sigma_D^2(X)]$

Saturated: a dummy per value of X , so D on the controls gives $e(X)$ exactly. Let $\tau(X) = E[\tau_i | X]$, $\sigma_D^2(X) = \text{Var}(D | X) = e(1 - e)$.

Step 1 (FWL). $\tilde{D} = D - e(X)$ has mean zero, so $\delta_{OLS} = \text{Cov}(Y, \tilde{D}) / \text{Var}(\tilde{D}) = E[Y(D - e)] / E[(D - e)^2]$.

Step 2 (within a cell). The denominator is $E[(D - e)^2 | X] = \sigma_D^2(X)$. For the numerator, substitute $E[Y | X] = e E[Y | D=1, X] + (1 - e) E[Y | D=0, X]$:

$$E[Y(D - e) | X] = e E[Y | D=1, X] - e E[Y | X] = e(1 - e) (E[Y | D=1, X] - E[Y | D=0, X]) = \sigma_D^2(X) \tau(X),$$

the last step by **unconfoundedness**. **Step 3.** Iterate expectations over X in numerator and denominator. ■

Weights depend on i only through X_i : largest where $e(X) = \frac{1}{2}$, **zero** where $e(X) \in \{0, 1\}$. No functional form for Y was used — saturation absorbed it.

Śłoczyński (2022). If X enters *linearly* instead, with Y^0, Y^1 linear in X : $\delta_{OLS} = w_1 \text{ATT} + (1 - w_1) \text{ATU}$, where

$$w_1 = \frac{(1 - \rho) V[p(X) | D=0]}{\rho V[p(X) | D=1] + (1 - \rho) V[p(X) | D=0]}, \quad \rho = \Pr(D=1), \quad p(X) = \text{linear projection of } D \text{ on } X.$$

Equal variances give $w_1 = 1 - \rho$: **the ATT gets the control share** — the smaller group gets the larger weight.

Regression adjustment

- Interact treatment with every covariate:

$$Y_i = \alpha + \delta D_i + \gamma_0 X_i + \lambda D_i X_i + \varepsilon_i, \quad \widehat{ATT} = \hat{\delta} + \hat{\lambda} \bar{X}_1, \quad \widehat{ATE} = \hat{\delta} + \hat{\lambda} \bar{X}.$$

- Or center first: interact with $(X_i - \bar{X}_1)$ and read the ATT off $\hat{\delta}$; with $(X_i - \bar{X})$ for the ATE . Same coefficients, different covariate distribution.
- Equivalently: fit Y on X separately in each arm. For the ATT , impute Y^0 for treated units from the control fit and average the gap over the treated. For the ATE , also impute Y^1 for controls from the treated fit and average over everyone. `teffects ra` does both and gets the SEs right.

3. The NSW Demonstration

The 1970s empirical crisis

- The methods in this lecture were not adopted because they were elegant. They were adopted after a crisis of confidence in observational work.
- Job-training programs in the 1970s were evaluated observationally, and the estimates were all over the place — depending, apparently, on the analyst.
- The **National Supported Work (NSW) Demonstration** was different: it *randomized* participation. So the experimental effect was known.
- That created a **benchmark**: could observational methods, applied to the same treated units, recover the experimental answer?

What LaLonde did

- LaLonde (1986) ran the decisive test: take the NSW experiment, **throw away the experimental control group**, and replace it with comparison groups built from survey data (CPS and PSID).
- He applied the standard observational toolkit and compared the results to the experimental benchmark of roughly **\$1,800** in annual earnings gains.
- The estimates ranged from a fraction of the benchmark to **the wrong sign**, depending on how the comparison group was built.
- The paper was devastating, and a major reason the profession moved toward IV, RDD and DiD.

Dehejia and Wahba: not so fast

- **Dehejia and Wahba (1999, 2002)** revisited LaLonde's data with propensity score methods.
- Their finding: the observational estimates *did* recover the experimental benchmark reasonably well — **once common support was imposed**.
- Much of LaLonde's failure was an **overlap** problem, not only confounding: the CPS and PSID groups held huge numbers of people utterly unlike NSW participants.
- Discard the non-comparable units and the methods perform far better.

Smith and Todd: not so fast either

- **Smith and Todd (2005)** re-examined the Dehejia–Wahba result and found it fragile.
- The success was **sensitive to the subsample** analysed and to the **propensity score specification** chosen. Reasonable alternative choices undid it.
- Their charge is uncomfortable and worth stating plainly: the specification that worked may have been found *because* the experimental answer was already known.
- You cannot run that search in real research: there is no benchmark to check against — that is the point of doing the study.

The honest summary

- Selection on observables **can** work — when overlap is genuine and the confounders are truly measured.
- Neither condition is verifiable in the setting where you need them:
 - ▶ Unconfoundedness is **untestable**, always.
 - ▶ Common support is **checkable**.
- Do the check you can do, and argue explicitly for the assumption you cannot.
- **A useful exercise:** the NSW data are freely available. Apply every method here and compare to the \$1,800 benchmark — the fastest way to build instincts about when these methods hold up.

Wrapping Up

Summary [Part 1]: the score and what it does

- **The propensity score** reduces X to one number. Balance given $e(X)$ is free; the theorem — conditioning on $e(X)$ suffices — needs unconfoundedness. It buys dimension reduction, never **identification**.
- **Judge the score by balance, not fit** — and balance means the whole distribution.
- **Plot the overlap**. Common support is the one assumption you can check. Trim if you must, and say how the estimand changed.
- **IPW** reweights into a pseudo-population: $1/\hat{e}$ and $1/(1 - \hat{e})$ for the *ATE*; $\hat{e}/(1 - \hat{e})$ on controls for the *ATT*. Normalize (Hájek), and watch the tails.
- **Propensity score matching** is the weakest use: two units can share a score with opposite covariates, so pruning can *increase* imbalance (King and Nielsen 2019).

Summary [Part 2]: regression, and the verdict

- **The trade.** Matching needs overlap; regression needs a correctly specified outcome model. Both still need unconfoundedness.
- **FWL.** OLS identifies δ only from the part of D that X cannot predict.
- **The canonical model** bundles unconfoundedness, linearity and constant effects. With heterogeneous effects it returns a variance-weighted average, heaviest where treatment is a coin flip — generally neither the *ATE* nor the *ATT* (Angrist 1998; Słoczyński 2022).
- **Regression adjustment** — interacting D with X — recovers both, but still leans on functional form.
- **NSW.** These methods work when overlap is real and the confounders are measured — and you can only ever verify the first.
- Next lecture: designs that *create* variation instead of assuming it away.

References

- Angrist, J. (1998). Estimating the Labor Market Impact of Voluntary Military Service Using Social Security Data on Military Applicants. *Econometrica* 66, 249–288.
- Angrist, J. and J.-S. Pischke (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Busso, M., J. DiNardo and J. McCrary (2014). New Evidence on the Finite Sample Properties of Propensity Score Reweighting and Matching Estimators. *Review of Economics and Statistics* 96, 885–897.
- Crump, R., V. J. Hotz, G. Imbens and O. Mitnik (2009). Dealing with Limited Overlap in Estimation of Average Treatment Effects. *Biometrika* 96, 187–199.
- Cunningham, S. *Causal Inference: The Remix*, Ch. 5 (§5.9 for regression).
- Dehejia, R. and S. Wahba (1999). Causal Effects in Nonexperimental Studies. *JASA* 94, 1053–1062.
- Dehejia, R. and S. Wahba (2002). Propensity Score-Matching Methods for Nonexperimental Causal Studies. *Review of Economics and Statistics* 84, 151–161.
- Frisch, R. and F. Waugh (1933). Partial Time Regressions as Compared with Individual Trends. *Econometrica* 1, 387–401.
- Hájek, J. (1971). Comment on Basu. In *Foundations of Statistical Inference*, Holt, Rinehart and Winston, 236.
- Horvitz, D. and D. Thompson (1952). A Generalization of Sampling Without Replacement from a Finite Universe. *JASA* 47, 663–685.
- Imai, K., G. King and E. Stuart (2008). Misunderstandings between Experimentalists and Observationalists about Causal Inference. *JRSS Series A* 171, 481–502.
- King, G. and R. Nielsen (2019). Why Propensity Scores Should Not Be Used for Matching. *Political Analysis* 27, 435–454.
- LaLonde, R. (1986). Evaluating the Econometric Evaluations of Training Programs with Experimental Data. *AER* 76, 604–620.
- Lovell, M. (1963). Seasonal Adjustment of Economic Time Series and Multiple Regression Analysis. *JASA* 58, 993–1010.
- Rosenbaum, P. and D. Rubin (1983). The Central Role of the Propensity Score in Observational Studies for Causal Effects. *Biometrika* 70, 41–55.
- Słoczyński, T. (2022). Interpreting OLS Estimands When Treatment Effects Are Heterogeneous: Smaller Groups Get Larger Weights. *Review of Economics and Statistics* 104(3), 501–509.
- Smith, J. and P. Todd (2005). Does Matching Overcome LaLonde's Critique of Nonexperimental Estimators? *Journal of Econometrics* 125, 305–353.